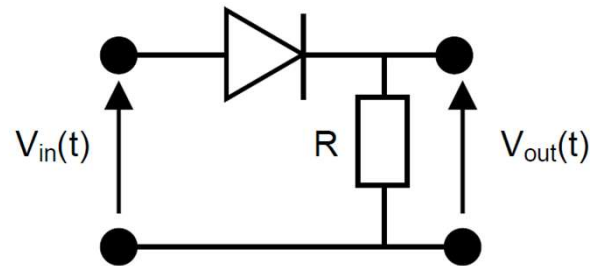




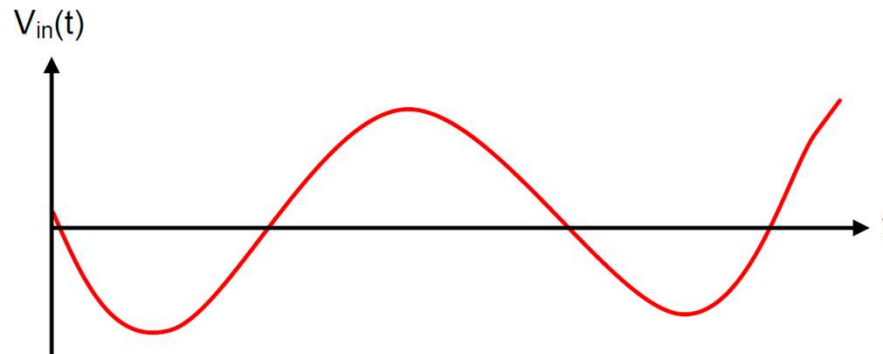
AC DC Converters

Half-Wave Rectifier

The primary function of a rectifier circuit is to convert an AC input voltage into a voltage that is either entirely positive or entirely negative, effectively eliminating the unwanted polarity of the input waveform. To illustrate this, let's consider the circuit shown below, which is known as a half-wave rectifier.



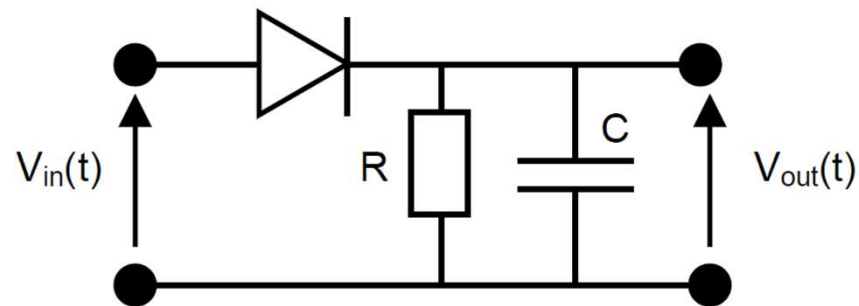
Q1) Determine the output signal $V_{out}(t)$ corresponding to the input signal $V_{in}(t)$ shown below.



Half-Wave Rectifier with Capacitor

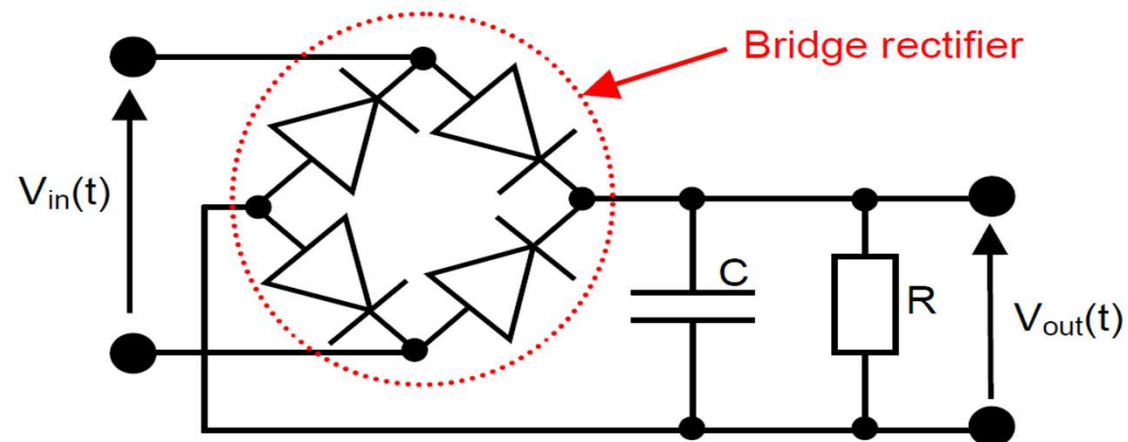
We can enhance the half-wave rectifier discussed in the previous question by adding a capacitor in parallel with the resistor. This modification results in a power supply circuit that converts an AC voltage input into a DC voltage output. Additionally, this circuit can be used as a demodulator in amplitude modulation (AM) communication systems.

Q2) Now, let's analyze the operation of the circuit shown below, assuming that the input signal is a sine wave.



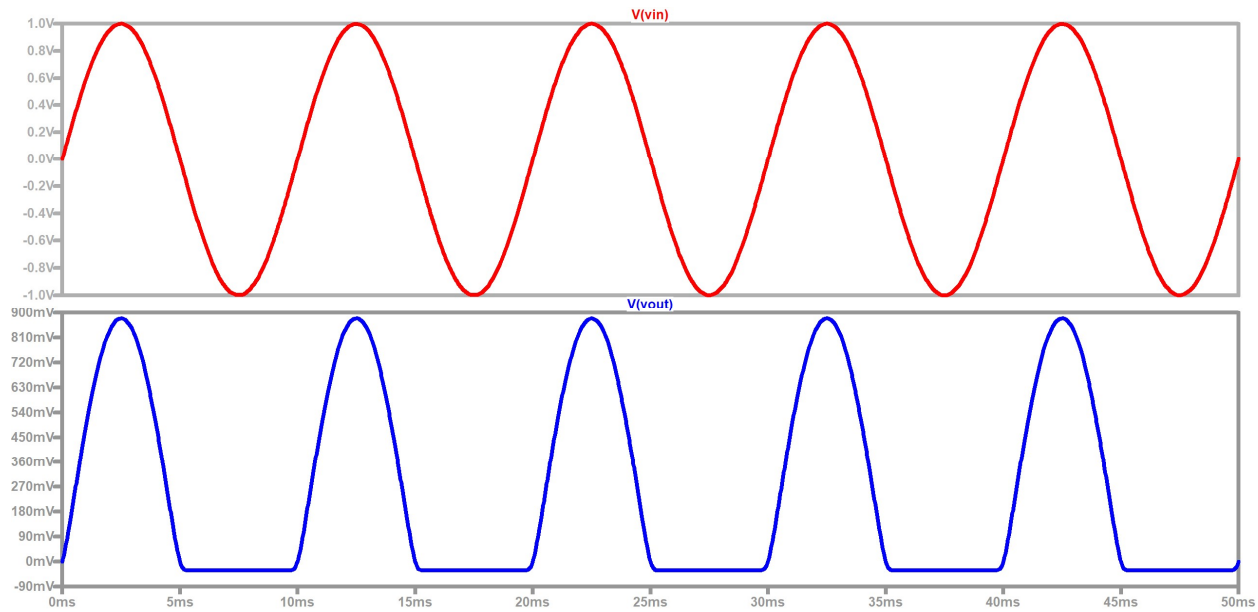
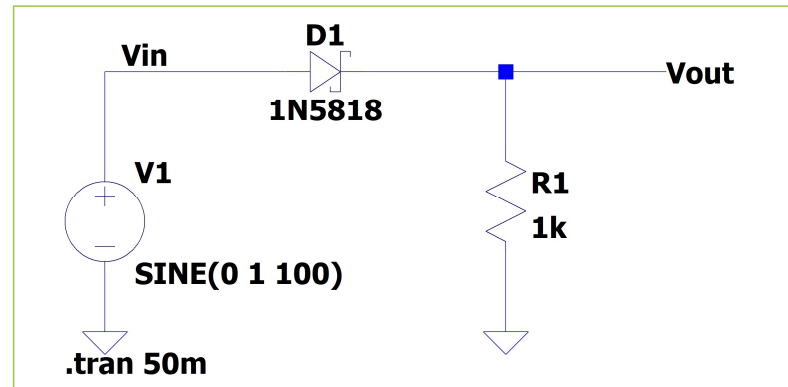
Power-Supply Circuit using a Full-Wave Bridge Rectifier

Q3) Analyze the operation of the circuit shown below, assuming the input signal is a sine wave.



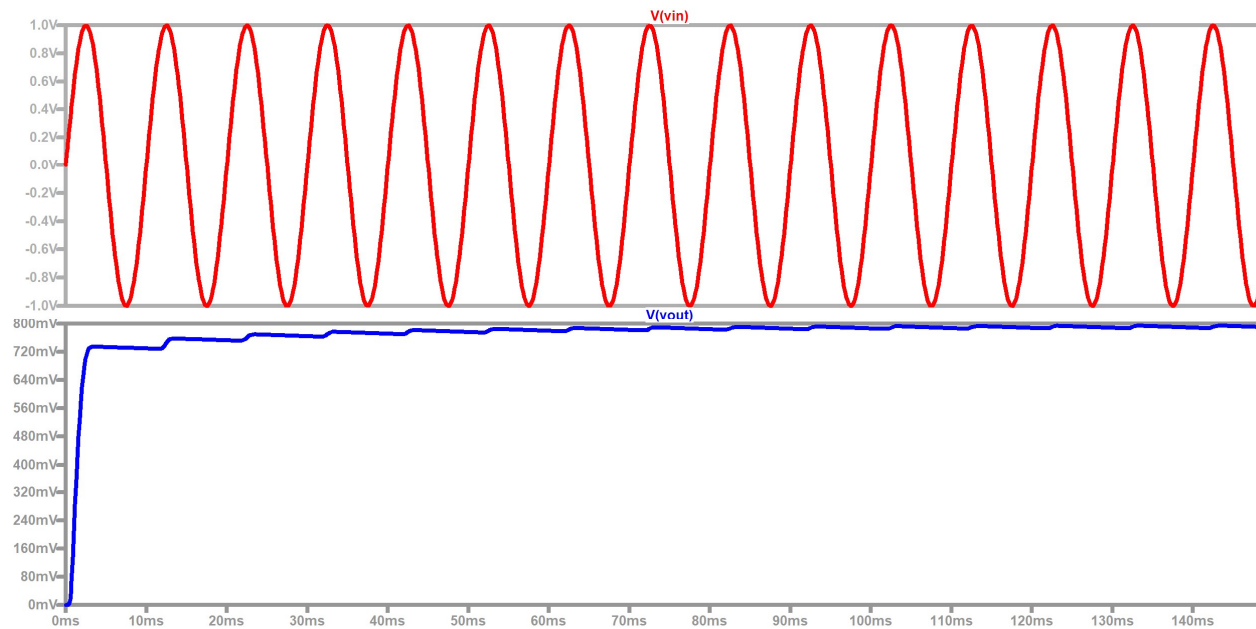
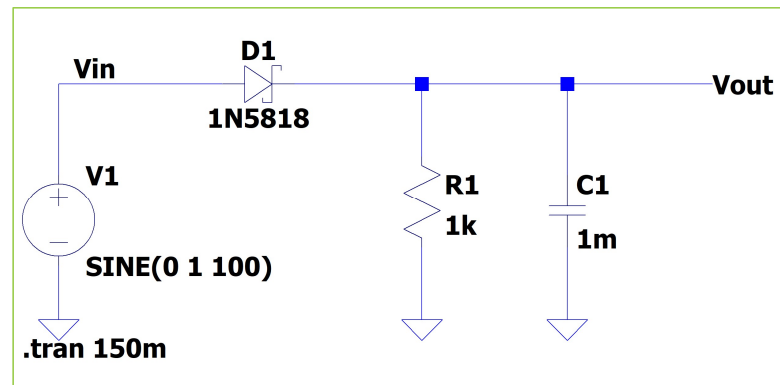
DIODE CIRCUITS

Half-Wave Rectifier



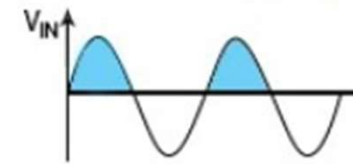
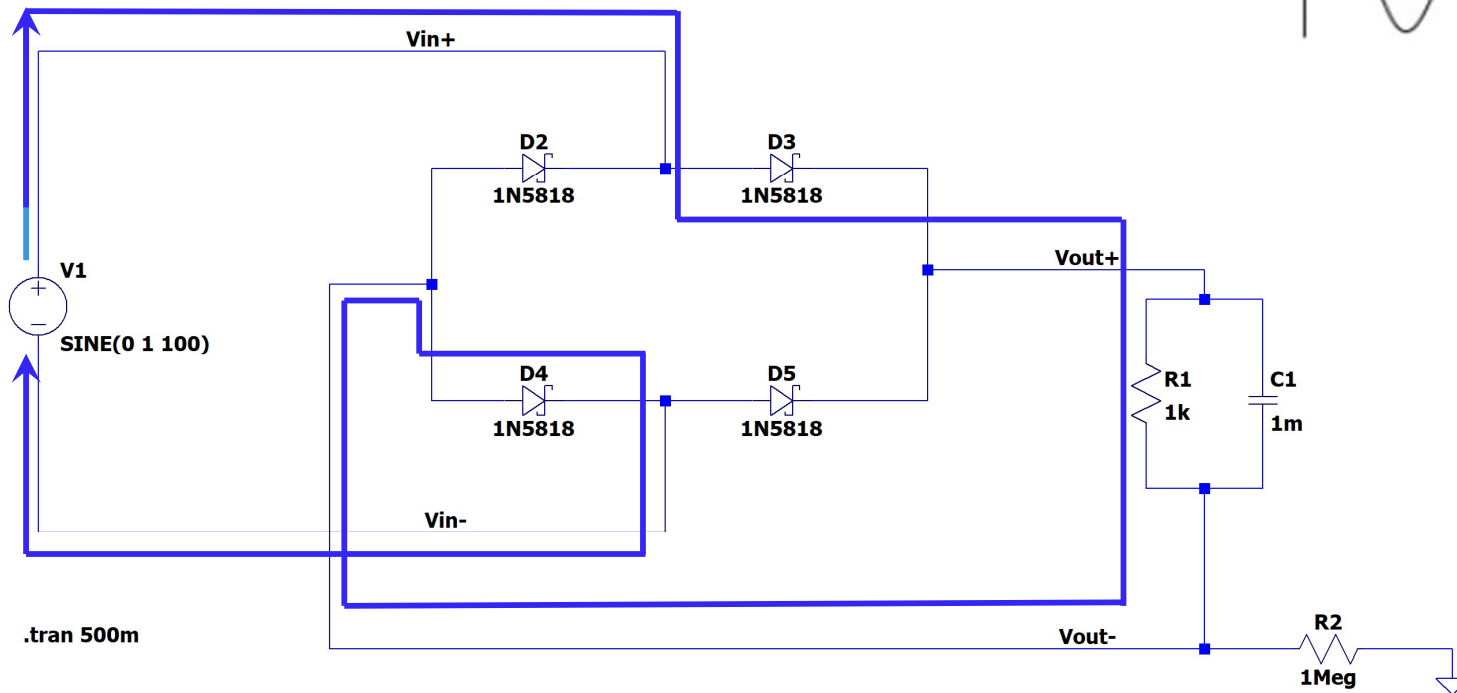
DIODE CIRCUITS

Half-Wave Rectifier with Capacitor



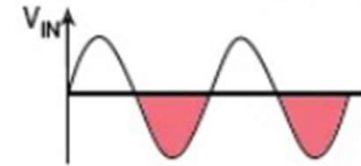
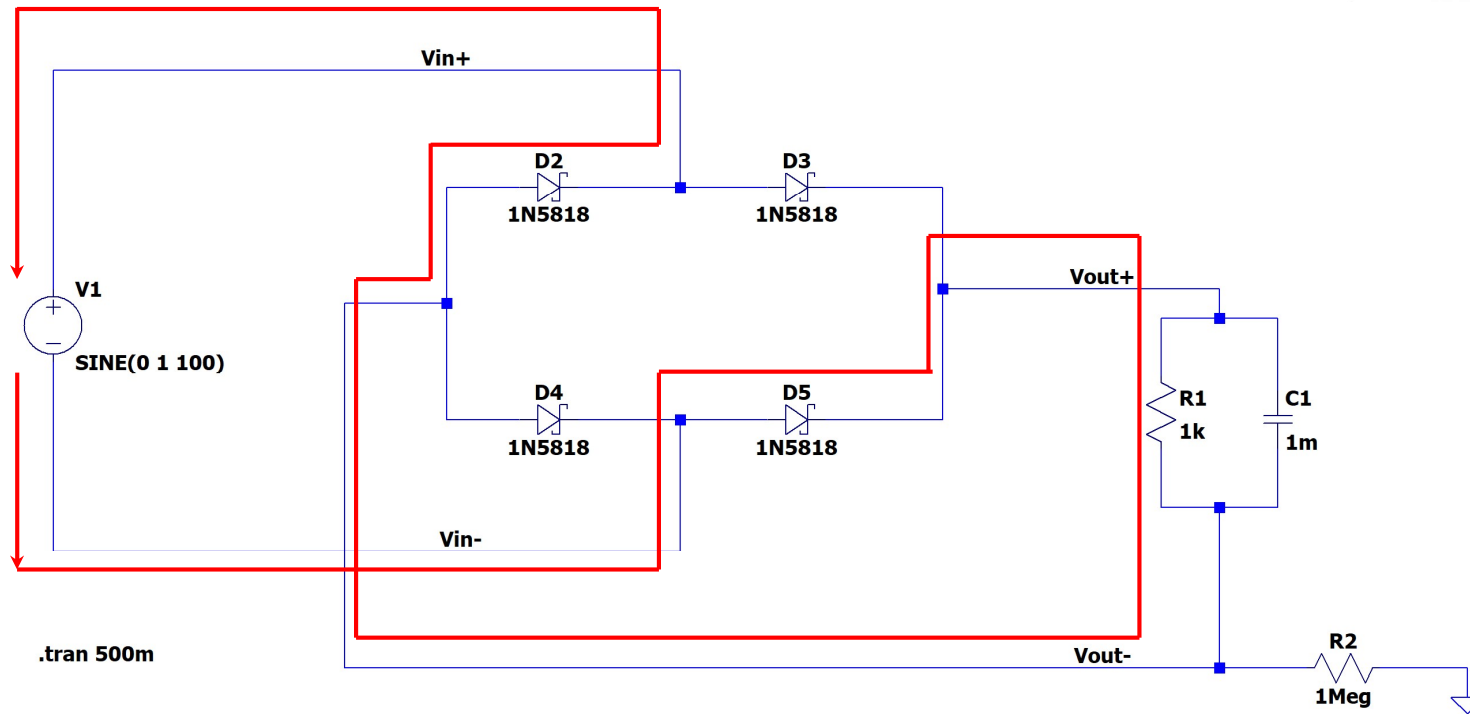
DIODE CIRCUITS

Power-Supply Circuit using a Full-Wave Bridge Rectifier



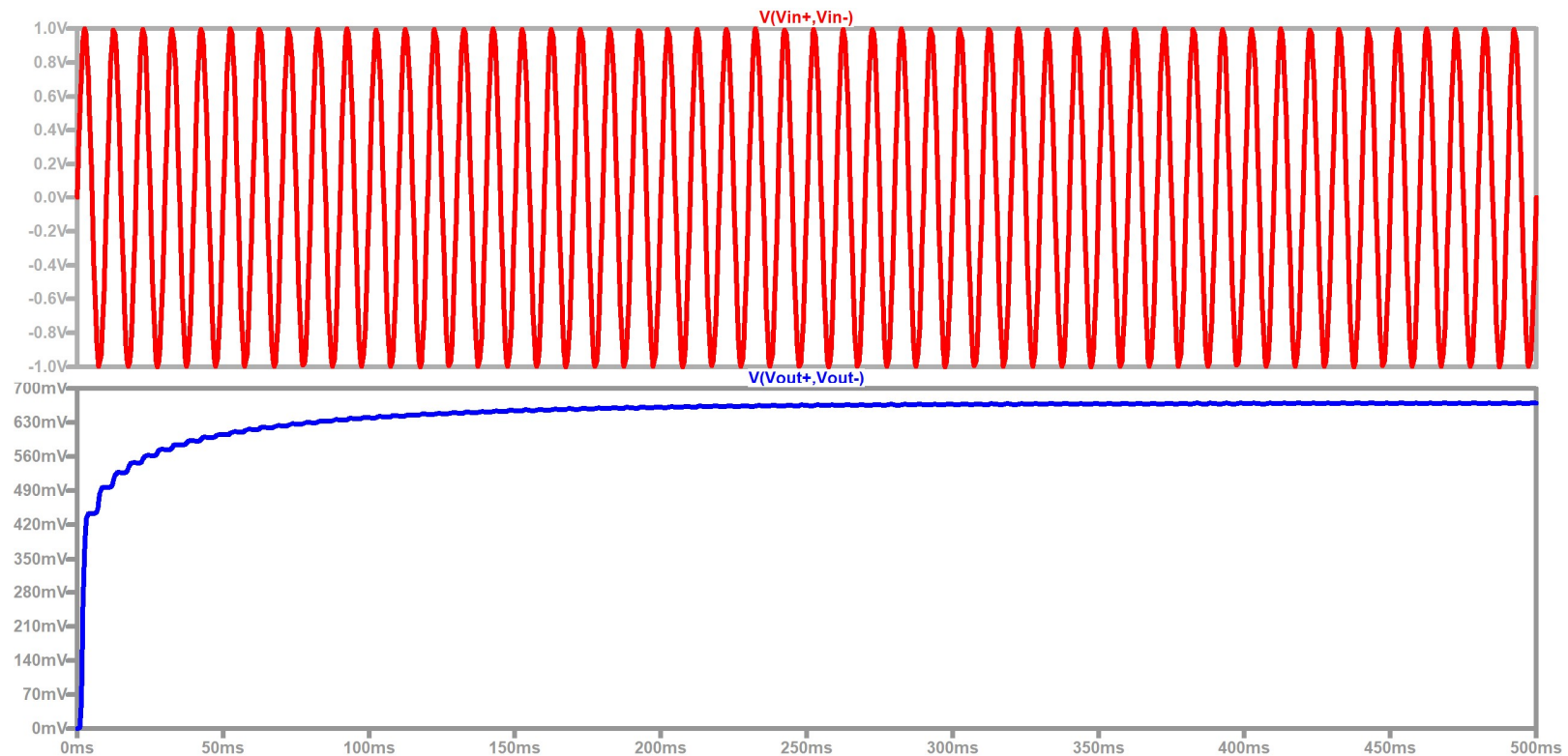
DIODE CIRCUITS

Power-Supply Circuit using a Full-Wave Bridge Rectifier



DIODE CIRCUITS

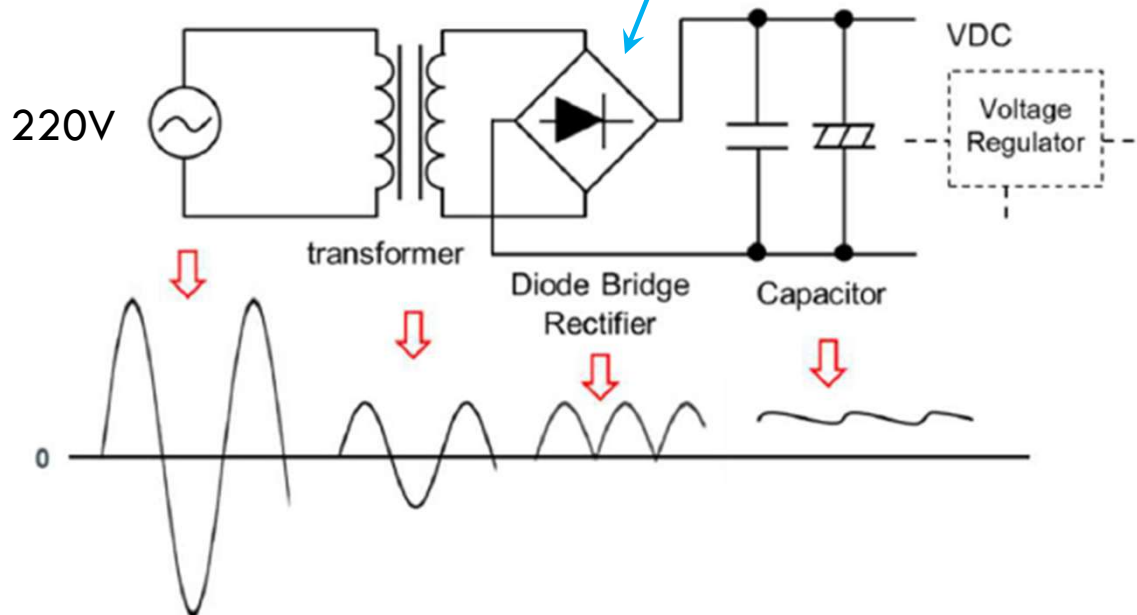
Power-Supply Circuit using a Full-Wave Bridge Rectifier



The output voltage is smoother compared with half-wave rectifier...

TRANSFORMER METHOD

Full-Wave Diode Rectifier

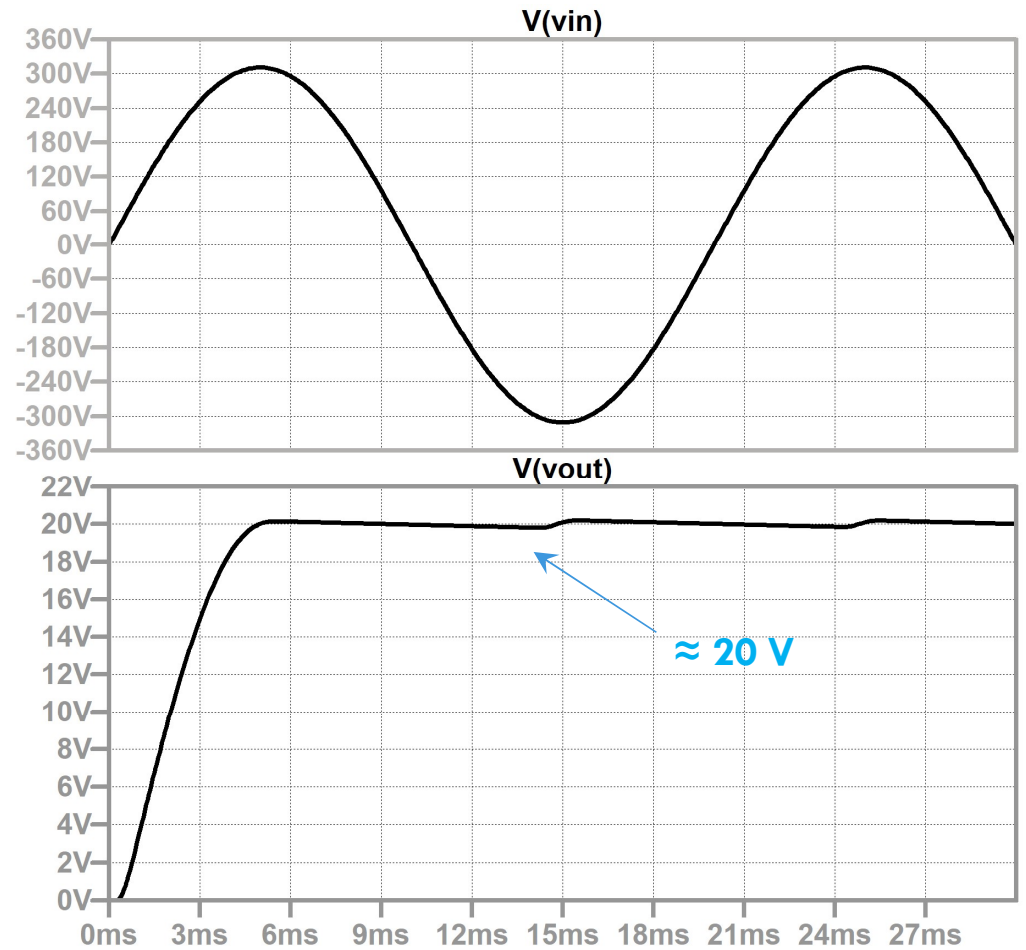
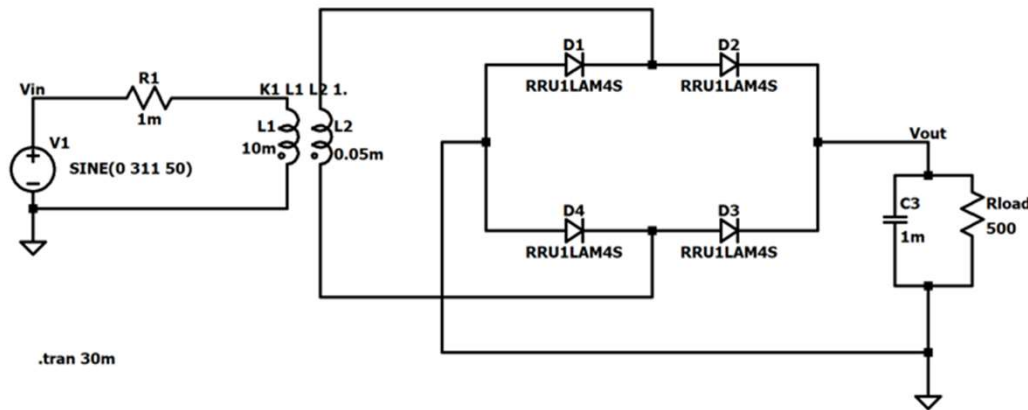


$$\frac{L_{\text{PRIMARY}}}{L_{\text{SECONDARY}}} = \left(\frac{N_{\text{PRIMARY}}}{N_{\text{SECONDARY}}} \right)^2$$

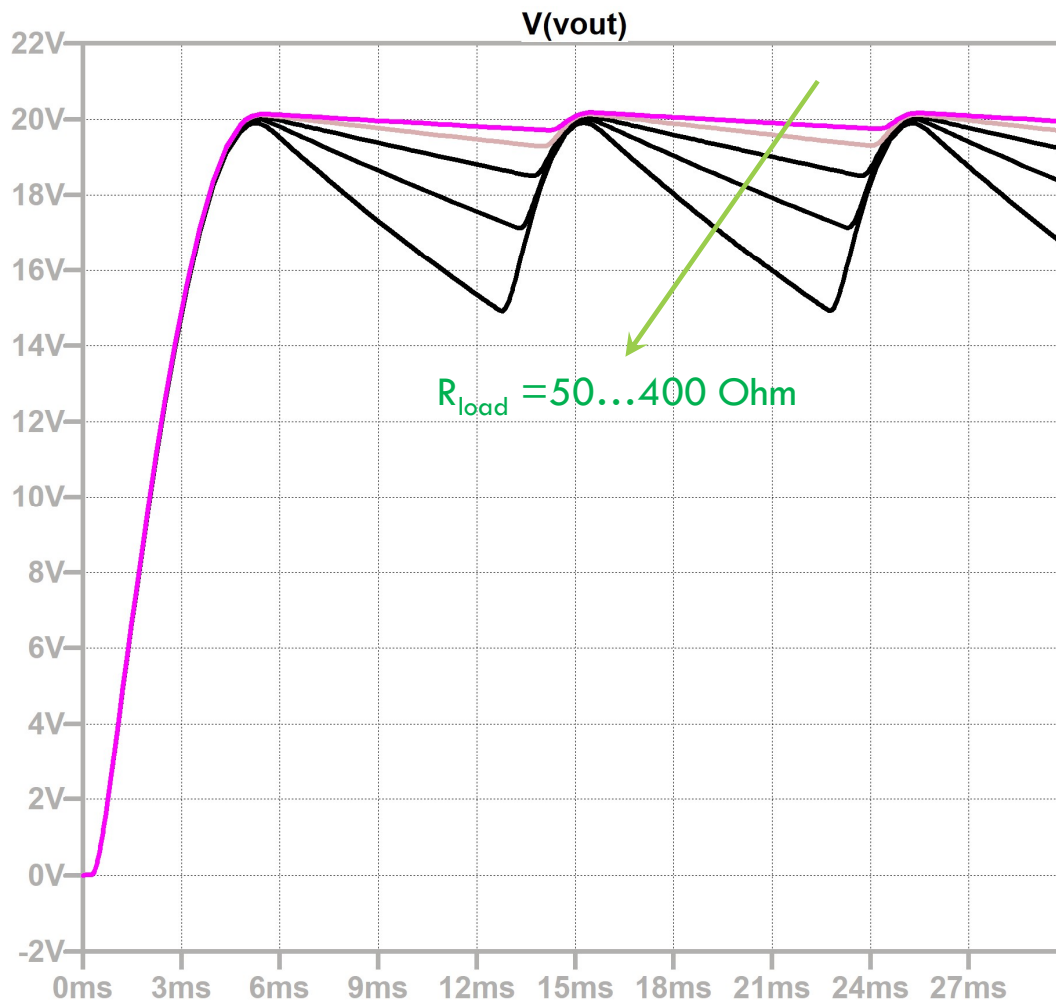
AC-DC Converter (220VAC Example)

- The input voltage is assumed to be 220VAC.
- This voltage is stepped down to a lower AC level using a transformer.
 - The transformer also provides isolation.
 - The winding ratio is chosen so that the DC output after rectification meets the desired value.
- The reduced AC voltage is converted to DC using a full-wave bridge rectifier.
 - A capacitor smooths the output, producing a DC voltage with small ripple.
 - The rectified DC voltage $\approx (AC \times \sqrt{2}) - \text{diode drop}$.
- If regulation is not required, this DC voltage can be used directly.
 - The voltage level depends on the transformer ratio.
 - As load current increases, the output voltage decreases.
- If regulation is required, a voltage regulator is added.
 - The secondary voltage of the transformer must be chosen to allow proper regulator operation.
 - Example: To obtain 12VDC output, the rectified voltage should be higher than 12 VDC (e.g. 14 VDC)
 - This margin ensures stable regulator function while keeping losses reasonable.

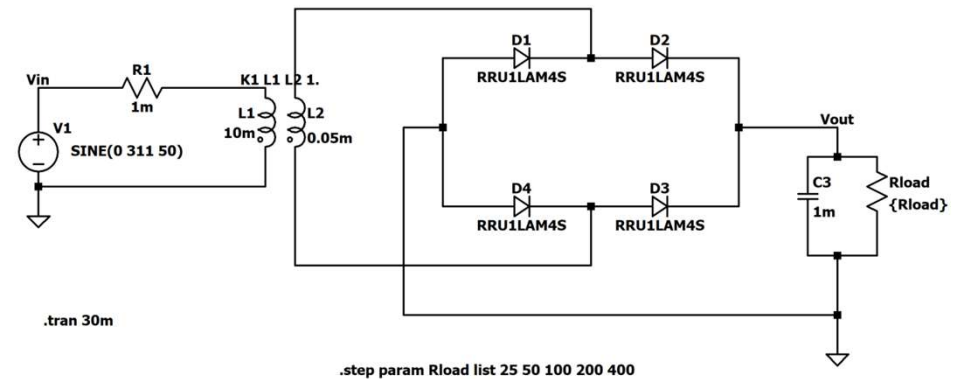
TRANSFORMER METHOD



TRANSFORMER METHOD

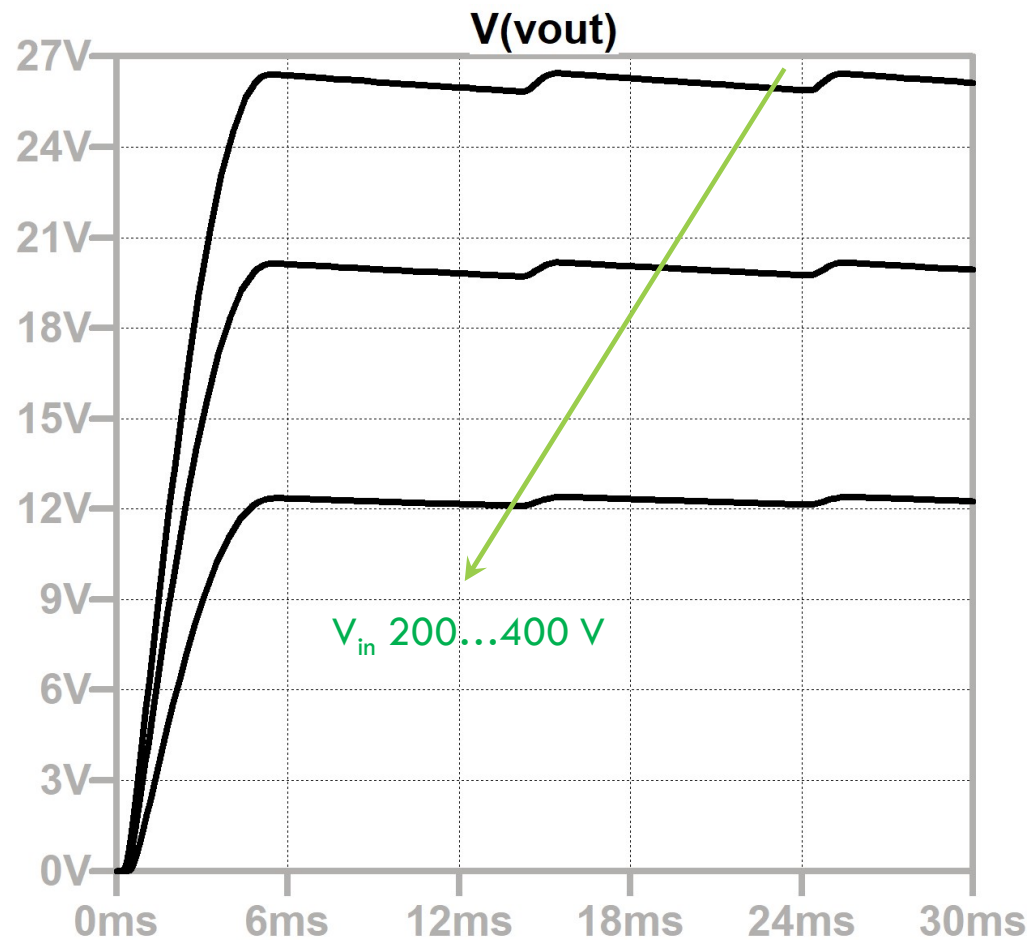


Output voltage regulation of transformer method is not satisfactory.



The smaller the load resistance, the more current has to be provided to the load! Higher Current $\rightarrow$ Larger Voltage Drop across Transformer Impedance, Diodes etc.

TRANSFORMER METHOD

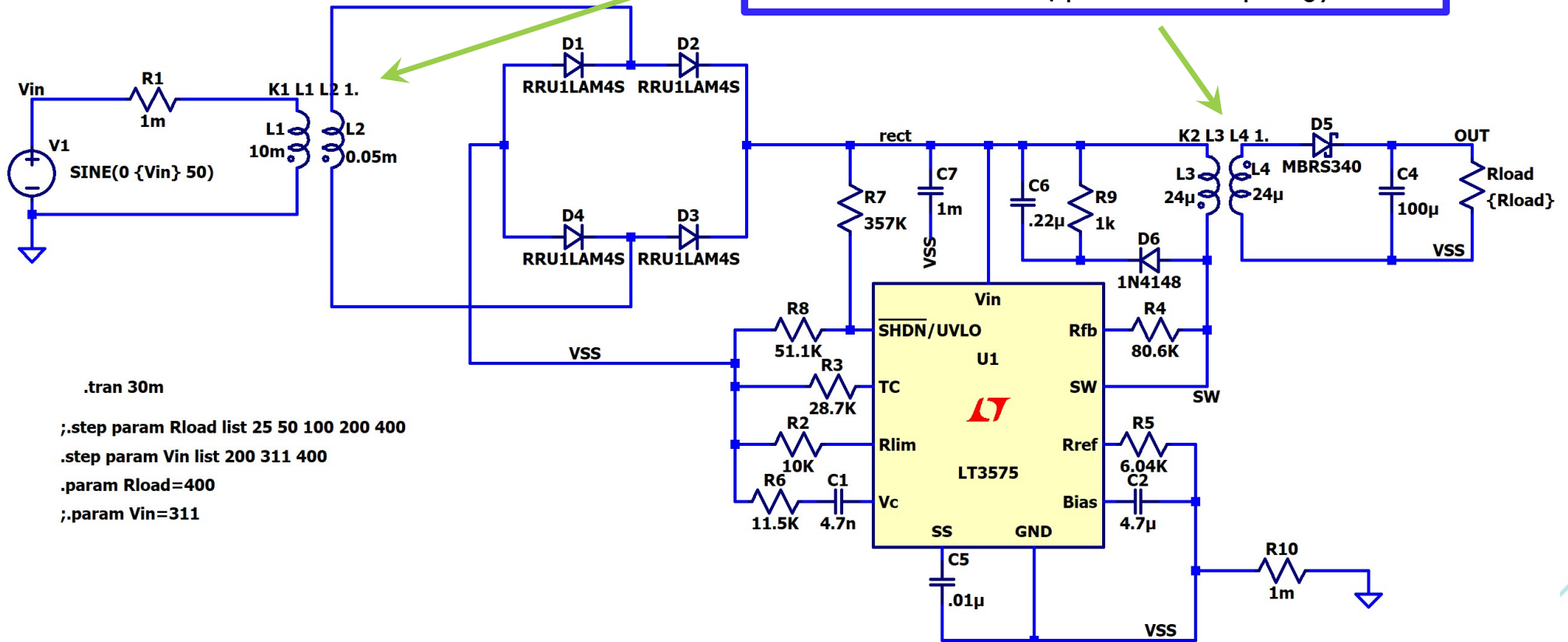


Input Voltage Variation:

If a constant output voltage is needed (e.g., 5V or 12 V), this is not possible with only Output voltage regulation of flyback method is not satisfactory.

FLYBACK CONVERTER

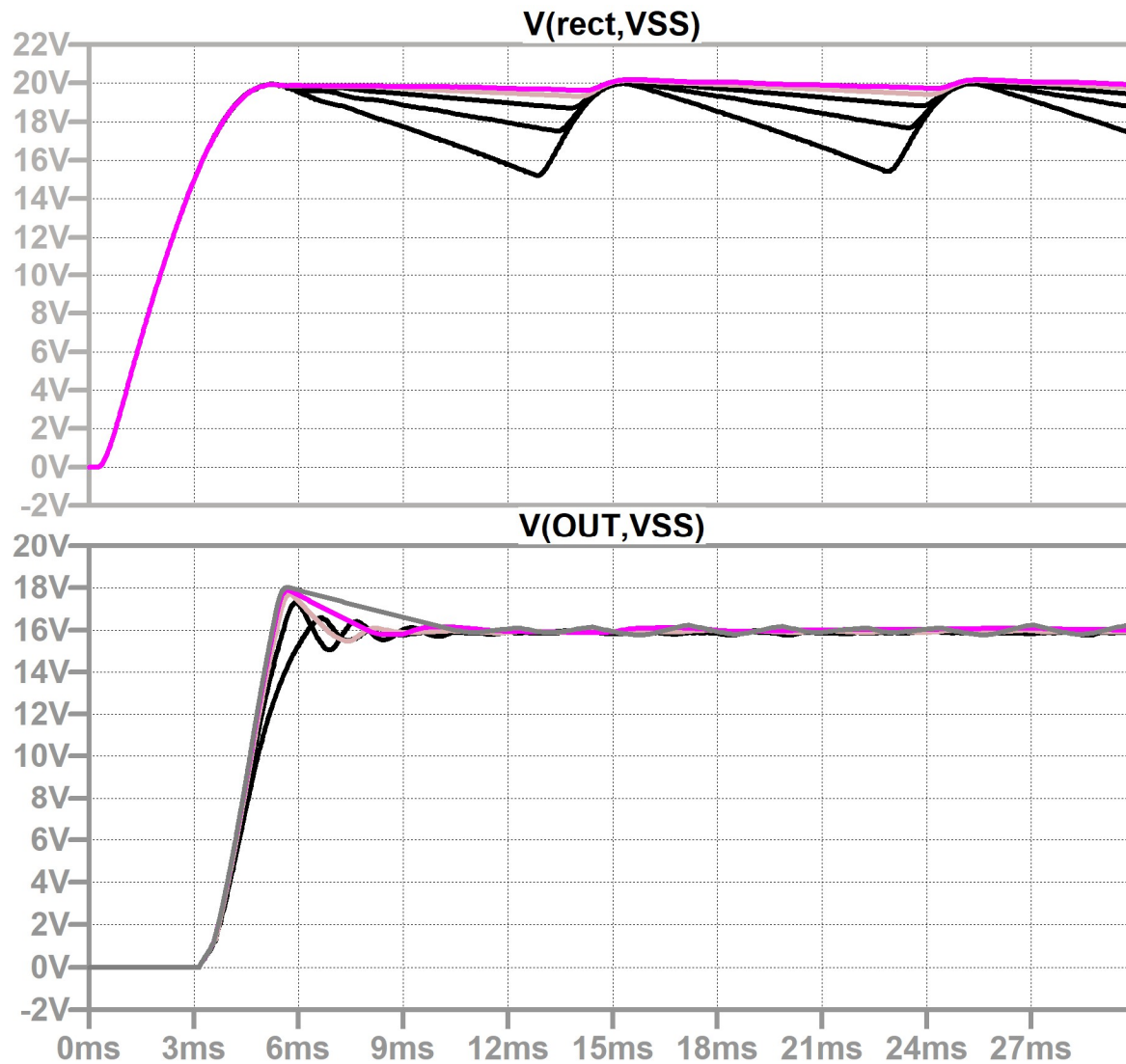
Two transformers – Alternative: the first transformer can be removed, please see Topology2



```
.tran 30m
;step param Rload list 25 50 100 200 400
;step param Vin list 200 311 400
.param Rload=400
;param Vin=311
```

TOPOLOGY 1

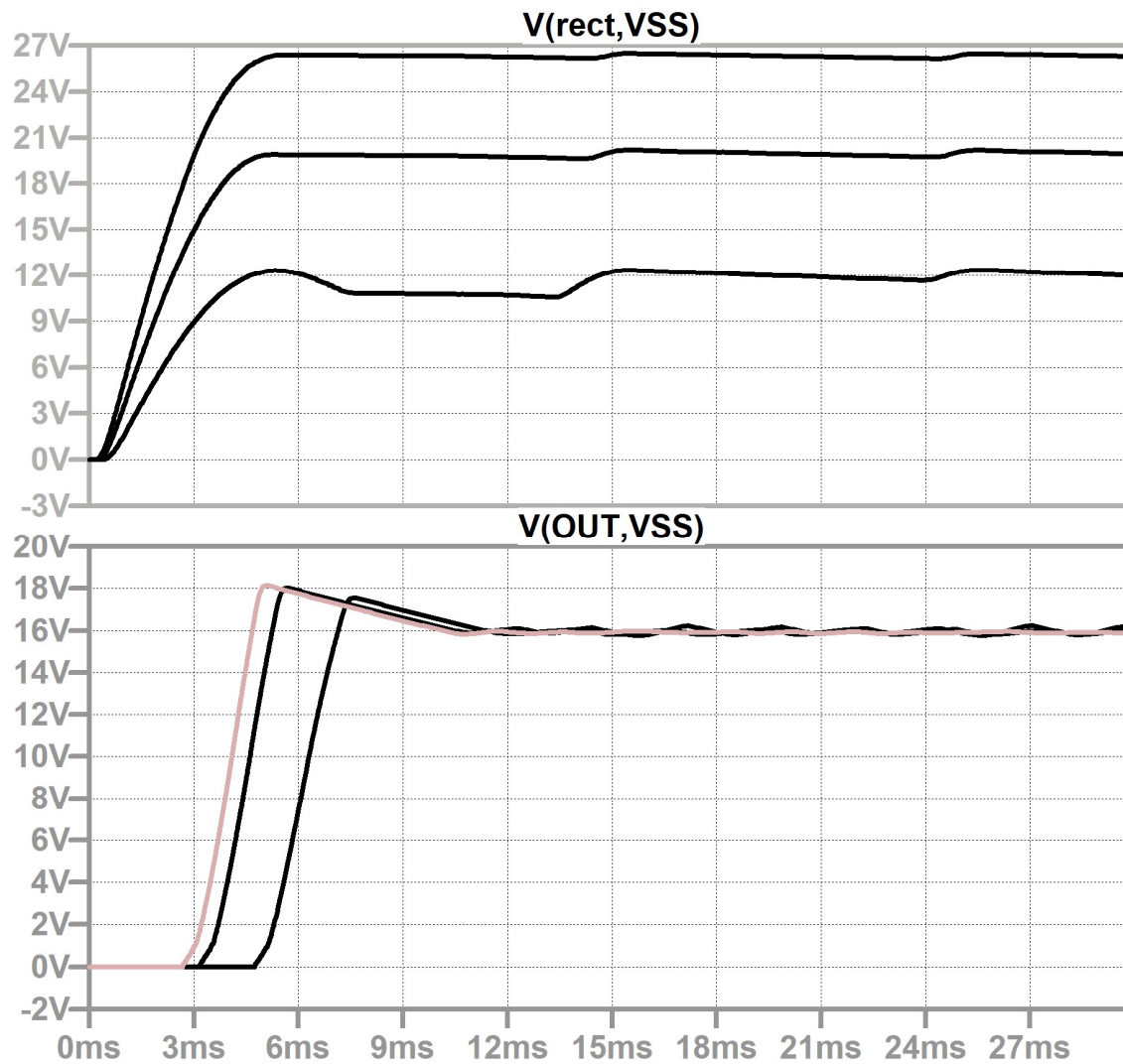
FLYBACK CONVERTER



Sweep the load resistance R_{load} :

`.step param Rload list 25 50 100 200 400`

FLYBACK CONVERTER

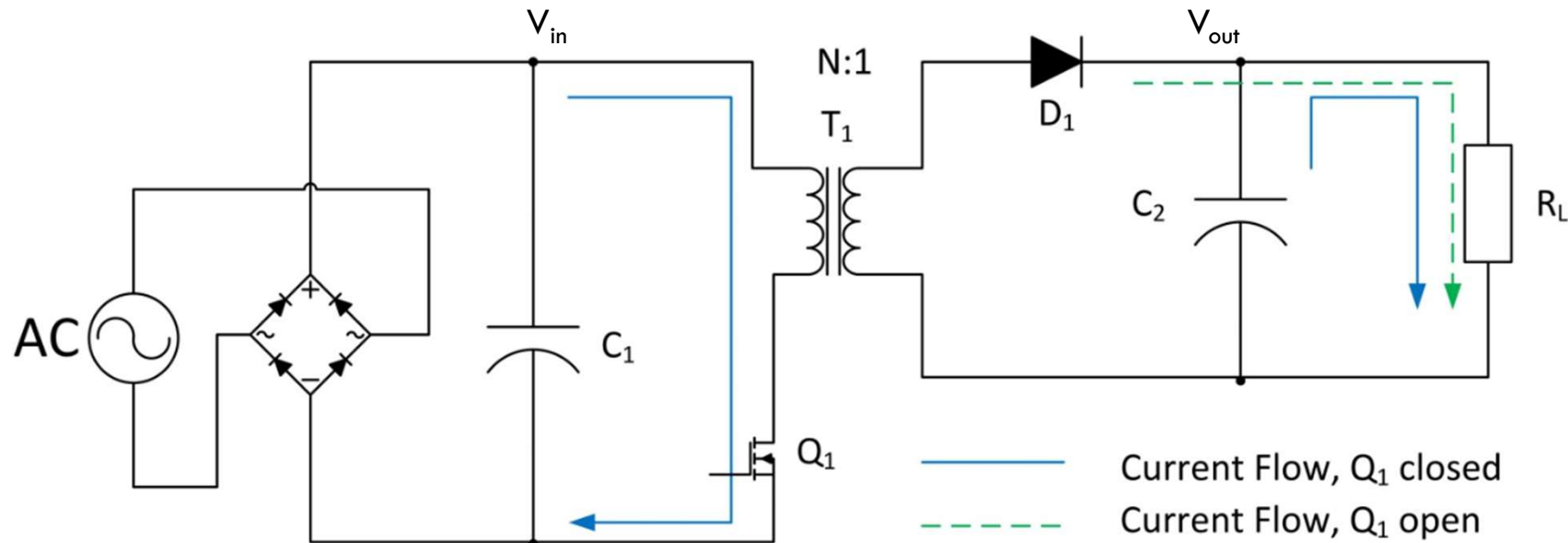


Sweep the input voltage V_{in} :

.step param Vin list 200 311 400

FLYBACK METHOD

TOPOLOGY 2 – please see the LTSpice Circuit example in the following slides



$$V_{out} = V_{in} \frac{1}{N} \frac{DC}{1 - DC}$$

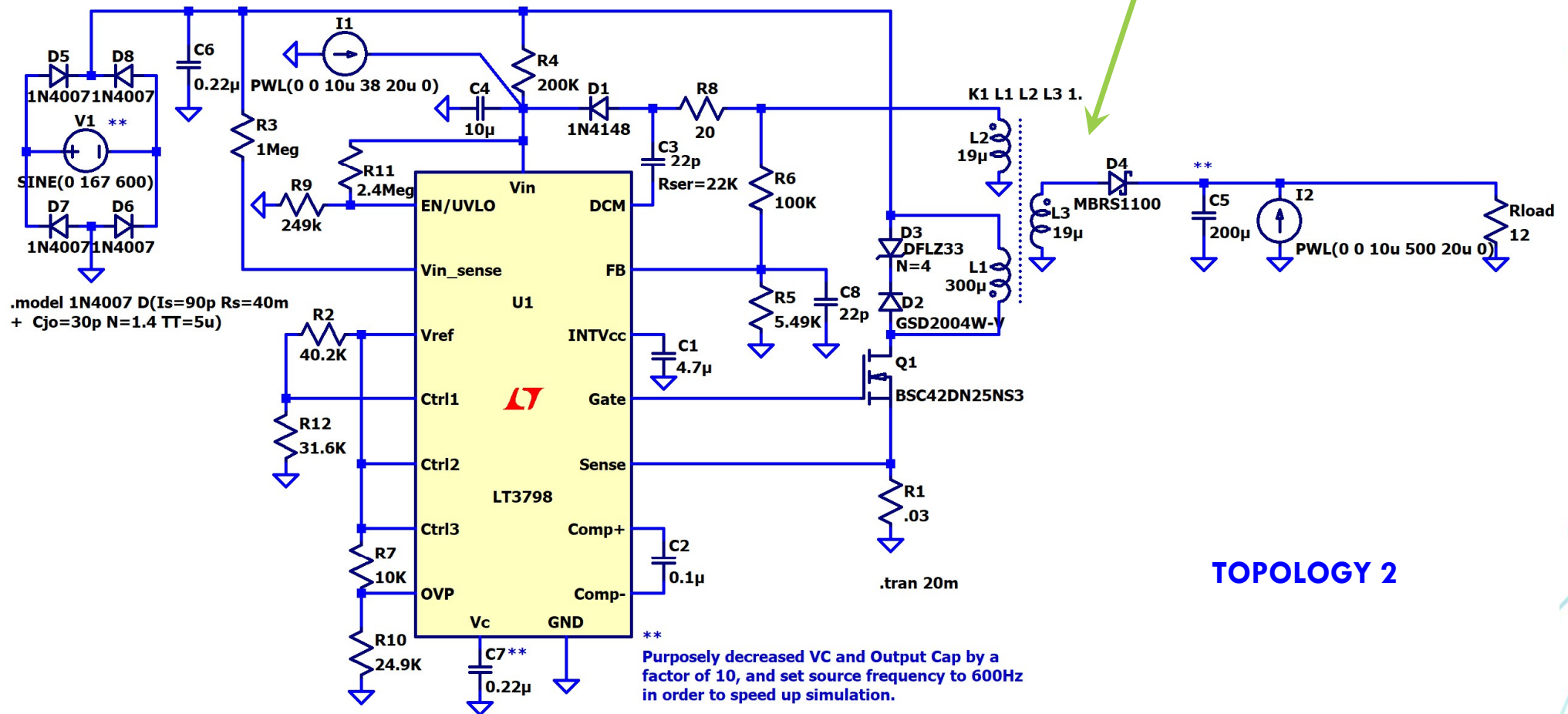
DC: Duty Cycle

N: Transformer Turn Ratio

High duty cycles (e.g., > 0.6) leads to dc winding losses and core saturation. Therefore duty cycles higher than 0.6 are usually not feasible → valid also for Flyback DCDC controllers!

FLYBACK CONVERTER

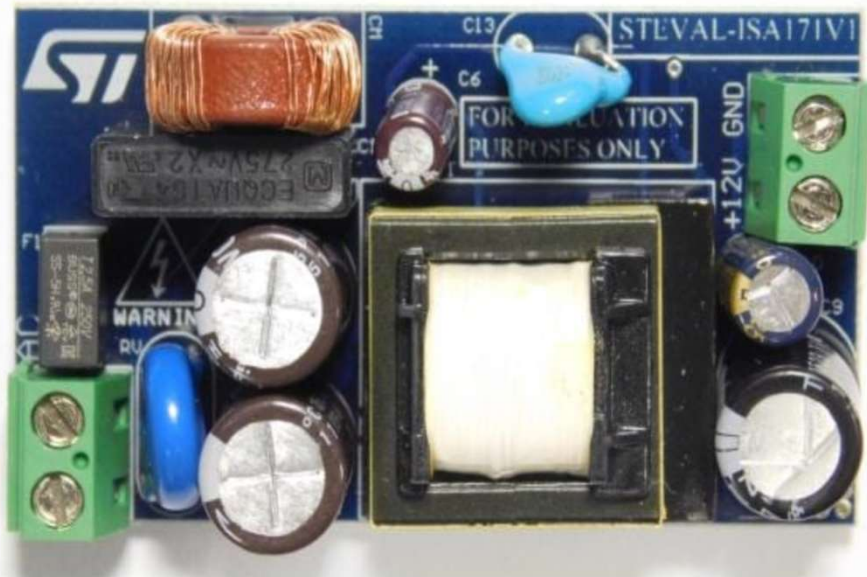
Single Transformer



TOPOLOGY 2

FLYBACK CONVERTER

Example



STEVAL-ISA171V1 reference design from STMicroelectronics

- Input power standby at 230Vac: < 30mW
- Wide input voltage range: 90- to 264-Vac
- Output voltage: 12V
- Output current: 1.25A
- Rated output power: 15W
- Active mode efficiency: >82%

POWER FACTOR CORRECTION (PFC)

A power factor correction (PFC) circuit reduces harmonic distortion in the supply current and shapes it closer to a pure sine wave, improving the power factor toward unity (1).

The power factor is defined as the ratio of active power consumed by a load (P_{act}) to the apparent power supplied (P_{apr}). Depending on whether the load is inductive or capacitive, the load current either lags or leads the voltage, increasing reactive power (P_{rea}). Many electronic devices operating under such conditions may generate harmonic currents or even cause stress to power transmission and distribution systems. Mathematically, PF (Power Factor) = $\cos\phi$, where ϕ is the phase angle between AC voltage and current. If the current waveform is highly distorted, PF drops well below 1.0. To restore the power factor closer to 1, harmonic current reduction is essential.

→ Please see Lecture Notes from EE 202 for Apparent and Active Power!

POWER FACTOR CORRECTION (PFC)

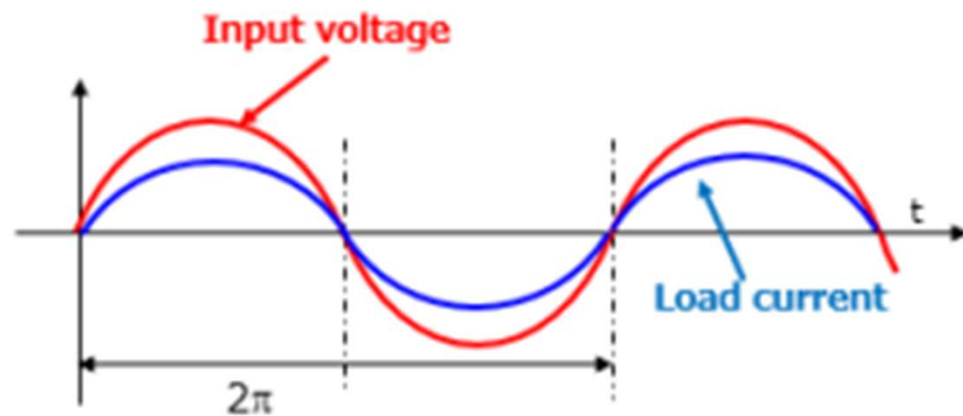
Resistive Loads

In the case of purely resistive loads, the supply voltage and the load current are in phase with each other ($\phi=0$). Hence, $\cos\phi = 1$. Therefore, electrical circuits with a purely resistive load have a power factor of 1.0. In this case, active power (P_{act}) is:

$$\begin{aligned} P_{act} &= \frac{1}{2\pi} \int_0^{2\pi} I_M \sin\theta \times V_M \sin\theta d\theta \\ &= \frac{1}{2} I_M \times V_M = \frac{I_M}{\sqrt{2}} \times \frac{V_M}{\sqrt{2}} = I_{rms} \times V_{rms} \end{aligned}$$

Apparent power (P_{apr}) is:

$$P_{apr} = I_{rms} \times V_{rms}$$



POWER FACTOR CORRECTION (PFC)

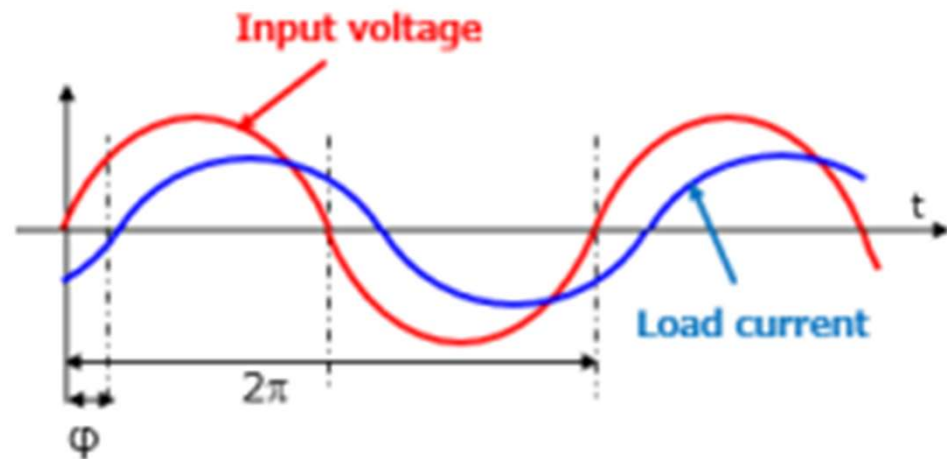
In the case of capacitive and inductive loads, there is a phase difference of ϕ degrees between the current and voltage waveforms. In this case, active power (P_{act}) is:

$$\begin{aligned} P_{act} &= \frac{1}{2\pi} \int_0^{2\pi} I_M \sin(\theta + \phi) \times V_M \sin\theta d\theta \\ &= I_{rms} \times V_{rms} \times \cos\phi \end{aligned}$$

Apparent power (P_{apr}) is:

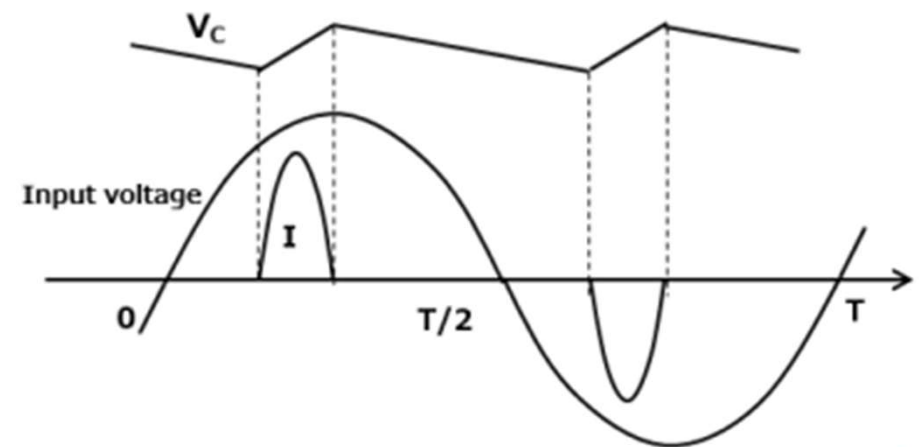
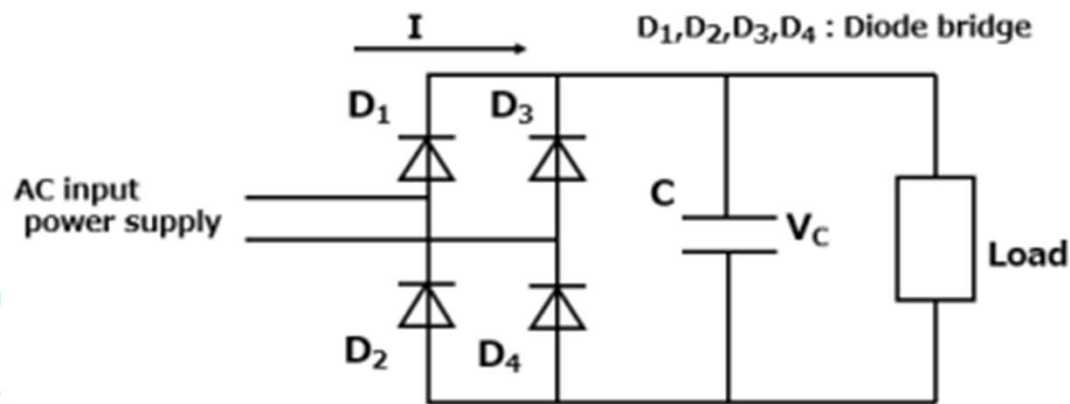
$$P_{apr} = I_{rms} \times V_{rms}$$

Hence, the power factor (PF) is calculated as $P_{act}/P_{apr} = \cos\phi$.



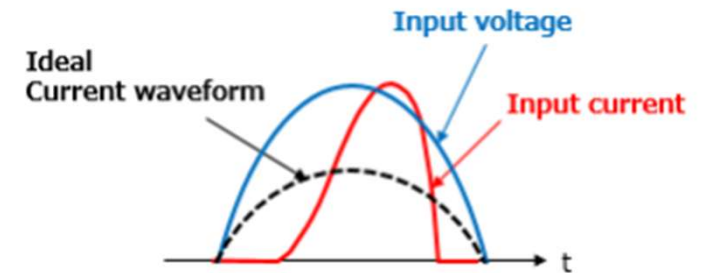
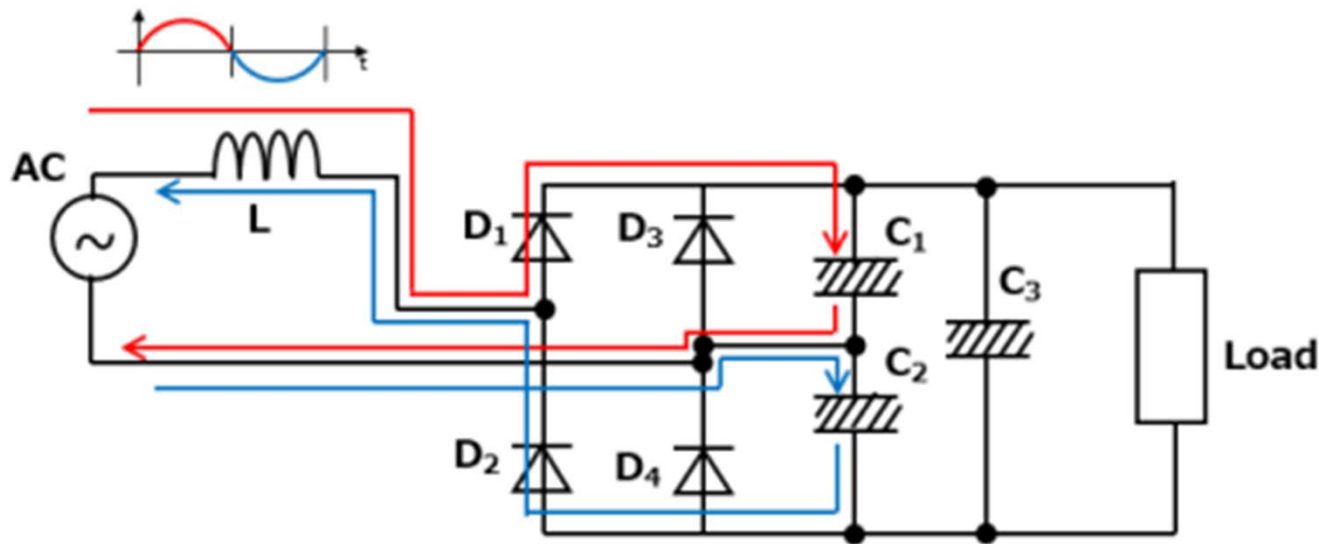
POWER FACTOR CORRECTION (PFC)

In a circuit using a smoothing capacitor, harmonic current is generated as the charging current for the capacitor distort from the input sine wave current. In the circuit below, the input AC voltage is applied to an input smoothing capacitor (C) via a diode bridge rectifier. Consequently, current (I) flows only during periods when the input AC voltage is higher than the charging voltage of C. The figure below shows an example of an input current (I) waveform, which has considerable distortion relative to a sine wave because of many harmonic components and therefore causes the power factor (PF) to drop well below 1.0. The power factor correction (PFC) circuit is reducing the harmonic content of the input current and thereby bringing the waveform close to a sine wave. As a result, the power factor increases to close to 1.0

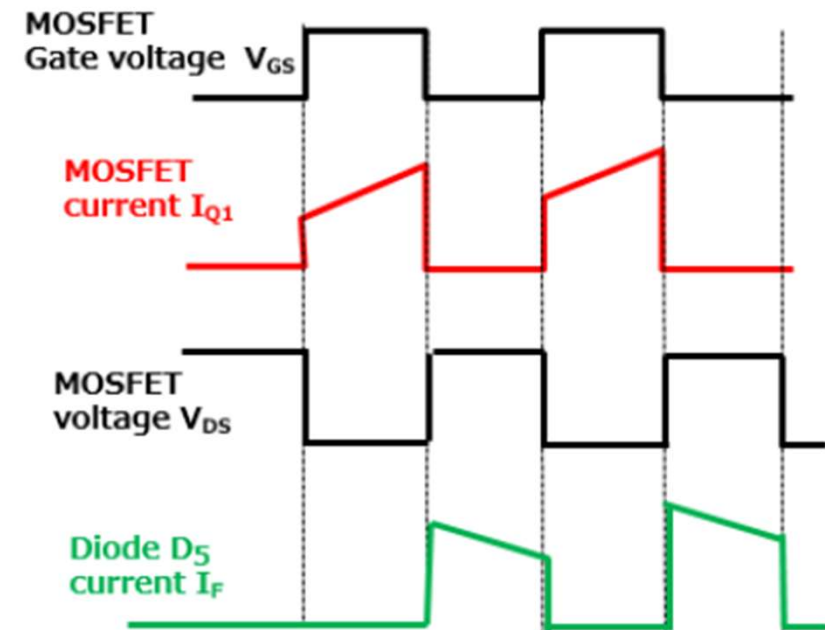
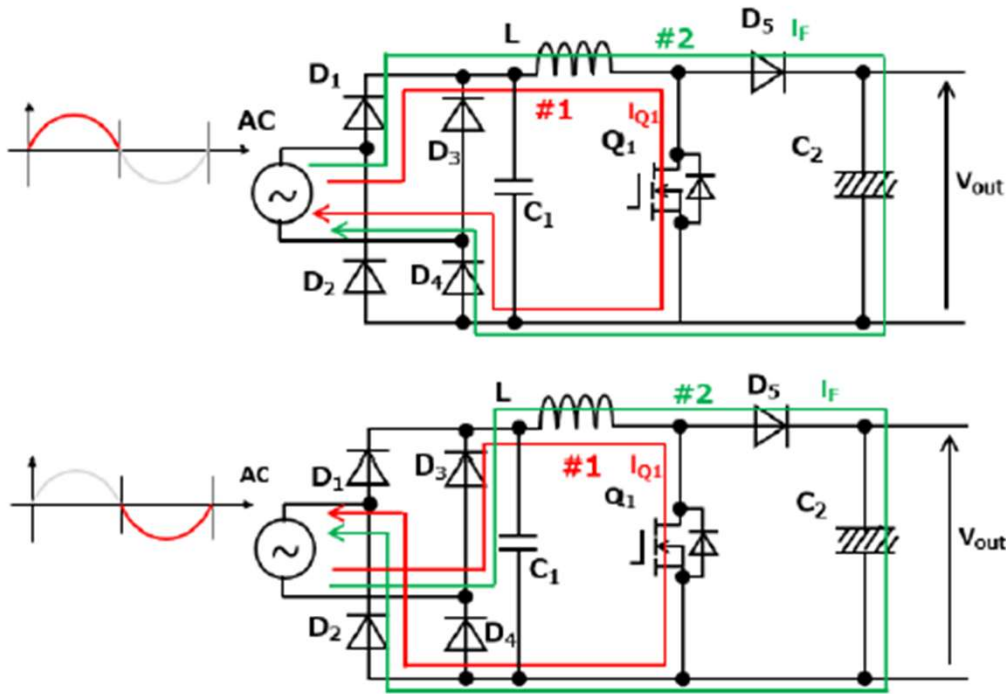


PASSIVE PFC

The figure below shows an example of a passive PFC circuit. The figure shows an example of an input current waveform for this circuit. The reactor (L) helps improve the power factor and the diodes and smoothing capacitors convert AC to DC. Because the passive PFC circuit operates at a mains frequency (50 or 60 Hz), a large reactor and capacitors are necessary. Therefore, passive PFC tends to be used for low-capacity power supplies.



ACTIVE PFC



Current path for MOSFET turn-on is indicated by arrow #1 whereas the one for MOSFET turn-off is highlighted by arrow #2.

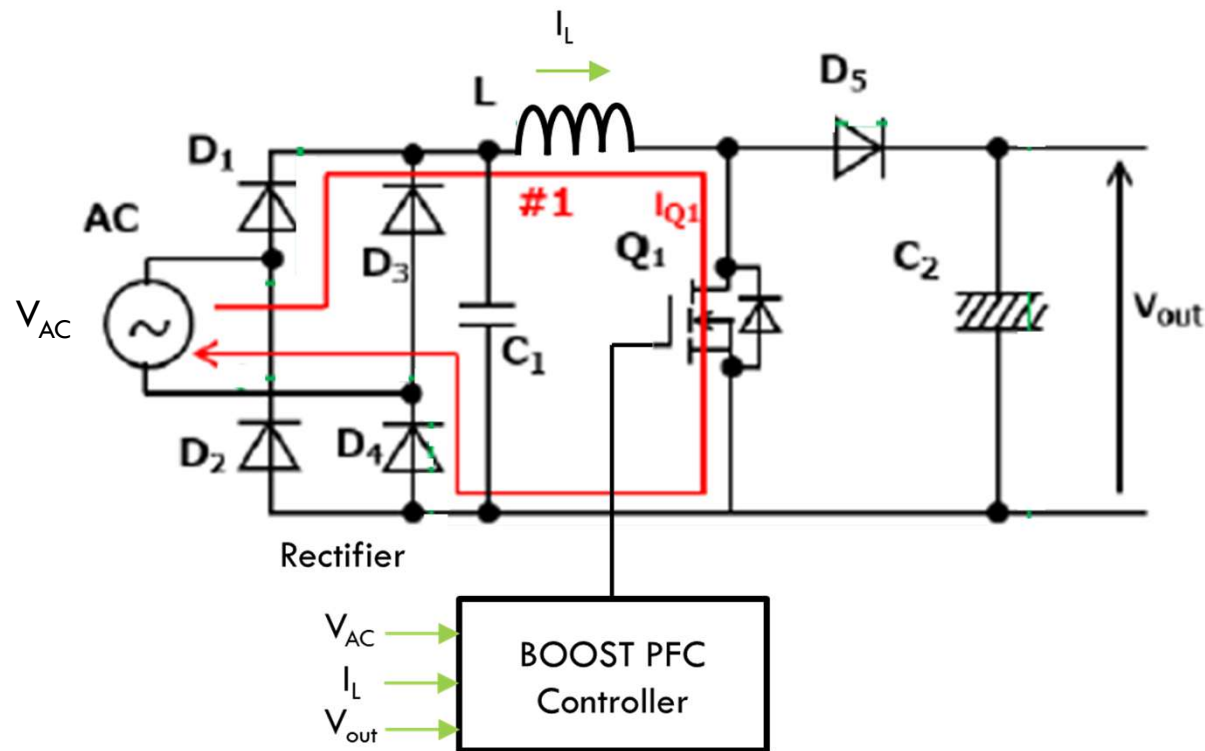
The Boost Converter IC shall support PFC → A boost converter IC regulates the output voltage to a level higher than the input by controlling the switch duty cycle. In PFC-capable versions, the IC additionally shapes the input current to follow the rectified input voltage waveform to achieve a high power factor.

ACTIVE PFC

A boost-based PFC circuit is a closed-loop control system whose goal is to keep the DC output voltage regulated while shaping the input current. To achieve this, the controller performs three key measurements:

- it measures the DC output voltage to keep it at the reference level,
- it measures the AC input voltage to generate a reference for the current, and
- it measures the average inductor current to ensure that this current follows the shape of the rectified AC voltage.

This coordinated control allows stable DC output voltage and near-unity power factor.



Example

Single-phase active Boost PFC converter is supplied from a 230 Vrms, 50 Hz AC mains. After an ideal diode bridge, the converter regulates the output to 400 V DC and delivers 1 kW to a resistive load. The converter operates in CCM at 65 kHz switching frequency. The inductor current ripple at the peak of the input voltage is specified as 20% of the peak input current at that instant. All components are ideal and neglect losses.

Input Voltage $V_{ac} = 230$ Vrms

Line Frequency $f_{line} = 50$ Hz

Output Voltage $V_0 = 400$ VDC

Output Power $P_0 = 1$ kW

Switching Frequency $f_s = 65$ kHz

Inductor Ripple $\Delta i_L = 20\%$ of peak input current

Operating Mode: CCM

a) Calculate the peak value of the AC input voltage

$$V_{pk} = \sqrt{2} \cdot V_{rms} = \sqrt{2} \cdot 230 = 325 \text{ V}$$

b) Determine the peak value of the input current at full load

$$P = V_{rms} \cdot I_{rms} \quad I_{rms} = \frac{P}{V_{rms}} \quad I_{rms} = \frac{1000}{230} = 4.35 \text{ A} \quad I_{pk} = \sqrt{2} \cdot 4.35 = 6.15 \text{ A}$$

Example

c) Calculate the duty cycle of the boost converter at the peak of the input voltage.

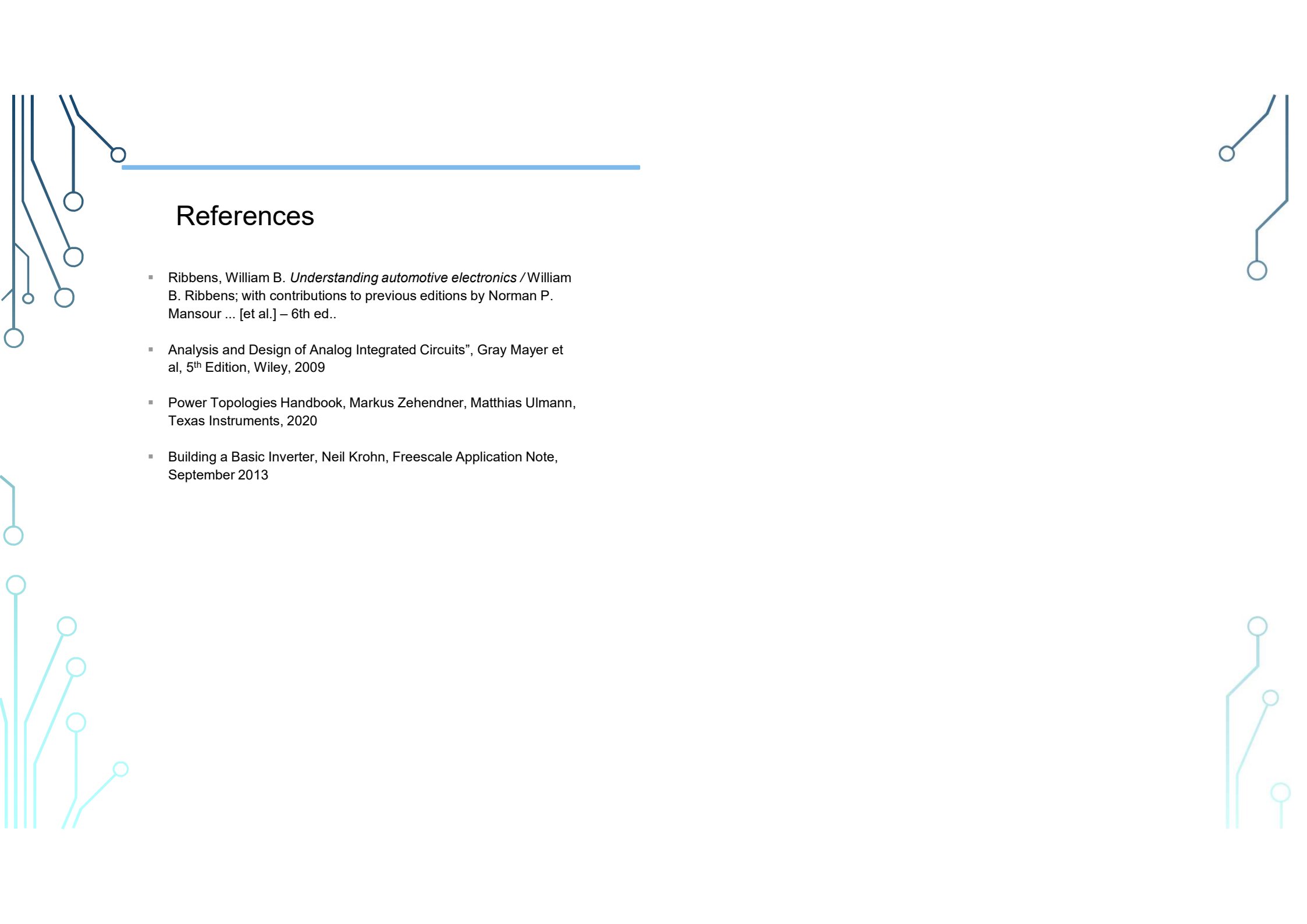
$$D = 1 - \frac{V_{in}}{V_o} = 1 - \frac{325}{400} = 0.1875 \approx 0.19 \text{ (19\%)}$$

d) Calculate the minimum boost inductance required to meet the ripple specification.

$$\Delta i_L = 0.2 \cdot I_{pk} = 0.2 \cdot 6.15 = 1.23 \text{ A}$$

For Continuous Conduction Mode (CCM):

$$\Delta i_L = \frac{V_{in} \cdot D}{L \cdot f_{sw}} \quad L = \frac{V_{in} \cdot D}{\Delta i_L \cdot f_{sw}} = \frac{325 \cdot 0.1875}{1.23 \cdot 65000} = 760 \mu\text{H}$$



References

- Ribbens, William B. *Understanding automotive electronics* / William B. Ribbens; with contributions to previous editions by Norman P. Mansour ... [et al.] – 6th ed..
- "Analysis and Design of Analog Integrated Circuits", Gray Mayer et al, 5th Edition, Wiley, 2009
- Power Topologies Handbook, Markus Zehendner, Matthias Ulmann, Texas Instruments, 2020
- Building a Basic Inverter, Neil Krohn, Freescale Application Note, September 2013